

# CAN VISION-LANGUAGE MODELS LEARN TO EXECUTE CLASSICAL SEARCH ALGORITHMS?

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Paper under double-blind review

## ABSTRACT

Prior work on LLM planning finds that operational reasoning over local actions improves with scale and chain-of-thought prompting, while structural enumeration of global state spaces remains comparatively flat. Existing training-based approaches supervise the output of a search algorithm, the final plan. We ask whether supervising the steps of a declared classical search algorithm can teach a vision-language model to execute it to completion. The model issues each search operation (expansion, successor generation, open-list update, or goal check) to a no-repair runtime, so replaying raw emissions reproduces the executed search. A random baseline samples valid operations at current states to measure learned control beyond mere validity. On a 15-task text suite under width-based best-first search, process supervision teaches execution with measurable advantage over this baseline, succeeding on 14/15 tasks versus 7/15 for random-valid and 0/15 for the pretrained base, with exact execution at 15/15 (gain 0.467, paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples)).

Across a broader 24-task, three-modality suite the trained policy never beats random-valid (125/288 versus 240/288), succeeding only under additive best-first algorithms. The dominant learned failure is a source-correct, applicable operation rejected by the search invariant, while the random-valid control fails late on budget exhaustion. Observation corruption localizes the learned controller in the language channel, where masking or shuffling collapses success (up to  $-0.944$ ), while blanking or degrading rendered states does not (at most  $-0.111$ ). Zero-shot transfer and reduced second-backbone replication show no reliable gains. All results are development-stage, with a single training seed per cell except two prospective replication cells (seeds 17/29/71, descriptive  $n=3$ ). The replicated headline cell (greedy best-first, multimodal) shows advantage over the base at every seed but never matches random-valid. The DAGger iteration-1 null also replicates. The BFWS development-panel positive remains single-seed. Within that scope, supervising search steps moves the two-competence account from observation to intervention. Learned search control is measurable only against baselines that already act validly, and lives in the textual scaffold rather than the rendered state.

## 1 INTRODUCTION

Classical planning requires finding a sequence of actions that transforms an initial state into a goal state under specified constraints, a search problem whose formal structure allows instances to be generated systematically, solutions verified automatically, and difficulty controlled precisely, making it a natural probe of reasoning in foundation models (Valmeekam et al., 2023b; Stein et al., 2025; Kokel et al., 2026). The formal model behind each instance fixes the task semantics, so errors reflect reasoning rather than ambiguity in the problem statement. The explicit state space also supports two levels of question: how actions apply locally, and how states connect globally. These properties let an evaluator vary difficulty along controlled dimensions and check each answer mechanically.

A recent study used this setting to measure latent planning competence in off-the-shelf language models (Huang et al., 2026). It separated two competencies that did not compensate for each other. Operational reasoning covers local applicability, progression, and validation: whether an

action is legal in a state, what state it produces, and whether a goal condition holds. Structural enumeration covers goal reachability, landmarks, and other global state-space structure. Across model scales, and under chain-of-thought prompting, targeted prompting, and scratchpad or agent harnesses, operational reasoning improved, but these gains were not accompanied by comparable structural gains. The evidence was item-level and text-only, drawn from models used as released, without task-specific training.

Structural enumeration is related to global search-space reasoning, but it is not identical to full search execution. Identifying which states are reachable, or which serve as landmarks, does not require controlling a frontier across a budgeted episode. The prior study established this profile observationally. It did not test whether the structural competency can be acquired through training, or whether a model can sustain a declared procedure over a complete episode. Closing that gap calls for direct intervention: train the model to make the exploration decisions of a classical search algorithm, then measure whether structural/process competence changes.

Our intervention replaces direct plan generation with learned execution of a declared search algorithm. A Search Process Policy (Section 3) is trained to emit Typed Search Operations, discrete steps such as expanding a state, generating a successor, updating the frontier, or applying a goal test. A Trusted Search Runtime validates and applies each emission, checks Algorithm Invariants on every operation, and maintains bounded external Search Memory holding frontier, visited, novelty, and priority state. This division is replay-determinative: the emission must contain every operand that determines exploration, and the runtime may not select omitted operands, reorder candidates for the policy, or silently repair an invalid emission. Replaying the raw emissions against the same task and Search Memory must reproduce the explored trace. Operation validity, invariant compliance, and episode success are measured separately from learned control advantage, which is computed against an oracle-assisted, validity-constrained random-valid control that samples only runtime-valid operations.

The study asks whether a policy can learn to execute declared algorithms with valid, invariant-compliant traces, whether learned execution yields control advantage over the random-valid control, and whether the acquired capability transfers to planning benchmarks and broader search-intensive reasoning. The active algorithms are BFS, a complete unpruned width-based best-first search (BFWS), and two additive best-first variants, weighted A\* (weight 3) and greedy best-first, both over  $h_{\text{add}}$ . Process supervision on search-trace segments is the core training condition, and two-iteration DAgger, model-generated successors, curriculum-by-modality, observation-corruption stress tests, and failure-mechanism calibration have also been executed. Modality conditions present the same task content as text-state, visual-state, or multimodal-state observations under equal Search Memory capacity. References are the pretrained base, the random-valid control, and exact classical execution (Section 4). Reduced-scope generalization/robustness and second-backbone stages, and full-scope zero-shot transfer to FOLIO, GSM8K, and HumanEval, have been executed under frozen protocols. Training-seed variance has been measured for two prospective replication cells (seeds 17/29/71, descriptive  $n=3$ ), with all other cells single-seed. The authorized final primary evaluation remains unmeasured [TODO: held-out final primary evaluation].

The results (Section 5) form a descriptive map of where learned control emerges, how it fails, and which observation channel carries it. On a frozen 15-task text-state development panel under BFWS, one fixed-seed process-SFT run achieved 0.933 success, versus 0.467 for random-valid, 0.0 for the base, and 1.0 for exact execution. It is the sole clear positive learned-control comparison in the study (the broader panels below return null or negative contrasts) and is development-stage.

On the BFS and additive best-first development panels, process SFT matched random-valid at 100%, so saturation left no room for measured advantage. An expanded baseline of 24 whole tasks held disjoint from all training rows across 12 domains and all three modalities sharpens the boundary: process SFT succeeded on 125 of 288 instances versus 240 for random-valid, 0 for the base, and 288 for exact execution, with every success in additive best-first cells, while BFS and width-based best-first failed on all 24 tasks in every modality. The development-panel BFWS positive does not reappear on this suite. A failure-mechanism calibration over 2,910 replay-verified episodes explains this pattern. The base fails immediately, predominantly on malformed outputs (379 malformed, 67 rejected frontier operations). The dominant learned failure is a parsed, source-correct, applicable operation rejected by the search invariant early in the episode. The random-valid control fails

only late on expansion-budget exhaustion. Process supervision teaches operation generation, not invariant-consistent frontier intent.

The executed extension arms bound this map. Learned adapters keep succeeding under 25 frozen semantics-preserving perturbation variants (128/150 versus 0/150 for the base, with the lossy name-compression stratum reported separately). Two DAGger iterations added no advantage over exposure-matched continued SFT (1/72 versus 1/72). Replacing trusted successors with model-generated ones dropped goal success from 45/45 to 1/45. Curriculum-by-modality showed no ordering-by-modality interaction, and zero-shot transfer to FOLIO, GSM8K, and HumanEval left no comparison surviving Holm correction (0/36).

A reduced second-backbone replication with InternVL3.5-8B (which shares the Qwen3-8B LLM family, isolating the vision tower, connector, and multimodal recipe) reached no goal in any modality. Observation-corruption probes then localize the learned controller: against 0.917 clean success, masking or shuffling the language channel drops success by up to 0.944 while blanking or degrading the rendered states drops success by at most 0.111, and the text-minus-visual isolation contrast is material ( $-0.694$ , 95% CI  $[-0.889, -0.472]$ ). The learned search control runs on the textual search-memory scaffold, and rendered visual semantics are near-dispensable under the frozen observation contract. All of these findings are development-stage, with a single training seed per cell except two prospective replication cells (seeds 17/29/71, descriptive  $n=3$ ) whose headline positive and DAGger null both replicate.

This paper makes three contributions.

- **From diagnosis to intervention.** We execute the training-based follow-up to a diagnosed structural bottleneck: rather than prompt around flat structural enumeration, we train models to execute declared search algorithms and map where learned control emerges.
- **A replay-determinative policy/runtime interface.** Typed Search Operations carry every exploration-determining operand, while the no-repair Trusted Search Runtime validates and applies them. Operation validity, invariant compliance, episode success, and learned control advantage remain separate measurements.
- **A descriptive map with failure and channel localization.** On the expanded baseline, learned success appears only in additive best-first cells, while the bounded BFWS development panel supplies the sole positive learned-control comparison. Learned failures localize early as source-correct, applicable operations rejected by search invariants rather than malformed outputs. Observation corruption shows the learned controller runs on the textual search scaffold rather than the rendered state. DAGger, model-generated successors, transfer, and second-backbone replication mark the negative boundaries. This is development-stage evidence, not final efficacy.

The paper therefore maps conditional, development-stage learnability of executable search rather than establishing general planning efficacy, with the held-out final primary evaluation, full-scope v1 generalization and second-backbone branches, and image-only state-history observation arms still unresolved [TODO: held-out final primary evaluation, full-scope v1, and image-only arms].

## 2 RELATED WORK

**Classical planning and search as LLM benchmarks.** A\* search (Hart et al., 1968) and its descendants define the family of heuristic best-first algorithms whose execution semantics our policy learns. Best-First Width Search (BFWS) combines goal-count exploitation with width-based exploration over a novelty measure (Lipovetzky & Geffner, 2017) and supplies the structural gate in our evaluation (Section 4). Recent benchmarks have tested whether LLMs can generate plans or reason about change in formal planning domains. PlanBench (Valmeekam et al., 2023a) found LLM plan generation far below classical solvers. A broader critical investigation showed that LLMs fail to plan even in simple domains where chain-of-thought prompting should suffice (Valmeekam et al., 2023b). AutoPlanBench (Stein et al., 2025) and ACPBench Hard (Kokel et al., 2026) extend these probes with automated prompt generation and harder reasoning tasks, confirming the gap. The observational study that motivates our intervention (Huang et al., 2026) diagnosed two non-compensatory competencies: operational reasoning improved with scale and prompting, but structural enumeration did not. All of these evaluate plans or plan-relevant judgments as end products. None asks the model to execute a declared search algorithm step by step.

**Neural algorithm execution.** Searchformer (Lehnert et al., 2024) is the nearest prior work in this thread. It trains a transformer to emit A\* search dynamics as serialized state add/remove tokens, then bootstraps to shorter-than-A\* traces on Sokoban. Searchformer operates on a single algorithm and domain family, serializes trace tokens without a typed-operation interface, and has no mechanism to validate or reject emissions during an episode. Stream of Search (Gandhi et al., 2024) serializes symbolic search processes, including backtracking and fruitless paths, into a unified language. A transformer pretrained on solver streams outperforms training on optimal trajectories alone. The model internalizes the full search process, but training and evaluation provide no per-operation validation or invariant checking. Earlier work established the broader thread. Veličković et al. (2020) trained GNNs with step-level imitation on individual BFS and Bellman–Ford steps, showing positive transfer across algorithms, though the setup used teacher-shaped per-node targets on synthetic graph encodings with no external runtime. Zaremba & Sutskever (2014) trained LSTMs to predict the output of short programs, showing that curriculum design was critical for length generalization. Our work differs from this thread in its typed operations, whose explicit operands fully determine exploration. A runtime validates each operation against algorithm-specific invariants during the episode. The evaluation reports operation validity, invariant compliance, and episode success as separate measurements.

**LLM/VLM planning and step-level supervision.** Several frameworks have the LLM participate in search by contributing local proposals while an external scaffold controls exploration. Tree of Thoughts (Yao et al., 2023) scaffolds tree-structured exploration with LM-based evaluation and backtracking, but the expansion and pruning decisions belong to the scaffold, not the model. RAP (Hao et al., 2023) places the LLM inside Monte Carlo Tree Search as both world model and reasoning agent, yet the UCT policy owns exploration. In our formulation, the model itself issues the operations that determine which states are expanded and which successors are generated.

Step-level and process supervision have shown advantages over outcome-only training. Scratchpads (Nye et al., 2021) train transformers to emit intermediate computation steps for algorithmic tasks, improving accuracy on long addition and program execution. Let’s Verify Step by Step (Lightman et al., 2024) demonstrated that per-step human labels outperform outcome supervision for mathematical reasoning at scale. These results support our choice of per-decision teacher labels. The distinction is that scratchpad tokens and human correctness judgments are unchecked free-form annotations. Our operations are typed and runtime-validated, bound to replay-verified positions in the search trace.

Recent work uses visual intermediate states in VLM reasoning. Visual Sketchpad (Hu et al., 2024) gives multimodal LMs drawing tools so reasoning proceeds over self-generated visual artifacts, improving mathematical and visual tasks. MVoT (Li et al., 2025) trains MLLMs to generate image visualizations of their reasoning traces. Our matched-modality arms serve a different purpose: they test which observation channel (text, rendered image, or both) carries the learned search controller, with corruption-based attribution (Section 5) rather than model-generated imagery.

**Runtime verification and constrained generation.** The LLM-Modulo framework (Kambhampati et al., 2024) couples LLM generation with an external verifier in a generate–critique–regenerate loop, arguing that LLMs cannot self-verify. The verifier feeds corrections back, so the loop participates in choosing the next attempt. Our Trusted Search Runtime does not repair: it validates, applies, or rejects each operation and charges the budget, without choosing omitted operands or reordering candidates for the policy. Self-Refine (Madaan et al., 2023) is the explicit single-model repair paradigm our runtime declines.

Constrained-decoding methods enforce structural well-formedness during generation. PI-CARD (Scholak et al., 2021) rejects inadmissible tokens at each decoding step, substantially improving text-to-SQL accuracy on Spider. Guided generation via FSM-indexed vocabulary (Willard & Louf, 2023) provides analogous guarantees model-agnostically. Both enforce syntactic structure without maintaining algorithmic state. Our runtime differs in kind. It maintains search-episode state (frontier, visited set, novelty counters, priority queue) and checks algorithm-specific invariants on every operation. Validity and invariant compliance are reported as separate episode-level measurements, not as a single accept/reject decision on a completed output string.

Table 1: Design-axis comparison of closest prior systems. **Algo. cond.:** policy conditioned on a declared algorithm’s step semantics. **Model expl.:** the model’s emissions determine exploration order. **No-repair RT:** the runtime rejects invalid operations without correction. **Inv. episodes:** full episodes checked against algorithm-specific invariants. **Match. mod.:** observations include non-textual modalities matched to the task. **Transfer:** cross-domain or cross-task generalization evaluated.

|                                             | Algo.                | Model      | No-repair  | Inv.                 | Match.     | Transfer             |
|---------------------------------------------|----------------------|------------|------------|----------------------|------------|----------------------|
| Searchformer (Lehnert et al., 2024)         | Partial <sup>a</sup> | Yes        | No         | No                   | No         | No                   |
| Stream of Search (Gandhi et al., 2024)      | Partial <sup>b</sup> | Yes        | No         | No                   | No         | Partial <sup>c</sup> |
| Neural execution (Veličković et al., 2020)  | Yes                  | No         | No         | No                   | No         | Yes                  |
| Process supervision (Lightman et al., 2024) | No                   | No         | No         | Partial <sup>d</sup> | No         | No                   |
| LLM-Modulo (Kambhampati et al., 2024)       | No                   | No         | No         | Partial <sup>e</sup> | No         | No                   |
| PICARD (Scholak et al., 2021)               | No                   | Yes        | Yes        | No                   | No         | No                   |
| Sketchpad (Hu et al., 2024)                 | No                   | Yes        | No         | No                   | Yes        | No                   |
| <b>Ours</b>                                 | <b>Yes</b>           | <b>Yes</b> | <b>Yes</b> | <b>Yes</b>           | <b>Yes</b> | Null <sup>f</sup>    |

<sup>a</sup>A\* dynamics serialized as tokens without typed operations or explicit operands. <sup>b</sup>Search streams serialized including mistakes without operation typing. <sup>c</sup>Self-improvement across search strategies, not cross-domain evaluation. <sup>d</sup>Per-step human labels for correctness, not algorithm-specific invariants. <sup>e</sup>External verifier provides correction feedback in a repair loop. <sup>f</sup>Zero-shot cross-task transfer evaluated with null result (Section 5).

Table 1 compares the closest prior systems on six design axes. Searchformer (Lehnert et al., 2024) is closest, training a transformer on A\* search dynamics, but it operates on a single algorithm without typed operations or runtime validation. Veličković et al. (2020) provide step-level algorithm imitation with cross-algorithm transfer but no runtime that checks operations during execution. PICARD (Scholak et al., 2021) rejects inadmissible outputs during decoding but enforces syntax on a single output string, not algorithm invariants over a stateful episode. No prior system in the table combines all six properties. This work combines algorithm-conditioned typed operations, a no-repair runtime with per-operation invariant checking, matched-modality observations, and cross-task transfer evaluation in a single system. Sections 4–5 report where this combination succeeds and where it does not.

### 3 SEARCH PROCESS POLICY

The introduction described the Search Process Policy mechanism informally. This section states the execution contract it runs under: the task inputs, the Modality Observation presented to the policy, the Typed Search Operation envelope, the Search Memory held by the runtime, the Algorithm Invariants checked on every operation, the ownership rule separating model decisions from runtime bookkeeping, and the learning objective. Training conditions, panels, and budgets are protocol decisions and appear in Section 4. Concrete operation schemas, invariant definitions, and runtime pseudocode appear in Appendix A.

A task instance is a triple  $x = (D, s_0, G)$ , where  $D$  is a formal domain model whose grounded actions induce a finite state space with deterministic transitions,  $s_0$  is the initial state, and  $G$  is a goal condition over states. An episode fixes a tuple  $(x, A, m, B)$ : an instance  $x$ , a declared search algorithm  $A$ , a modality adapter  $m$ , and a frozen budget  $B$  bounding the operations the episode may issue. The declared algorithm is part of the input, not a choice made at run time. The active algorithm set contains breadth-first search (BFS), the complete unpruned best-first width search (BFWS) with novelty and goal-count priorities (Lipovetzky & Geffner, 2017), and two additive best-first variants over the  $h_{\text{add}}$  heuristic (Bonet & Geffner, 2001): weighted A\* with weight  $w = 3$  (Hart et al., 1968; Pohl, 1970) and greedy best-first.

The policy never reads the instance directly. At decision step  $t$  it receives a Modality Observation  $o_t = \phi_m(c_t)$ , where the decision context  $c_t$  contains the canonical task context and the decision-relevant view of the current Search Memory, and  $\phi_m$  is a fixed observation adapter. Three adapters are declared. text-state serializes the state, the goal, and the search context as a compact relational representation. visual-state renders the same state and goal as a rendered state image plus a partial-goal constraint image. multimodal-state presents both. In all three adapters, Search Memory

contents and the grounded candidate list remain in the textual serialization. The adapters differ only in how the state and goal are presented. The adapter is fixed for an entire training or evaluation cell, and its contract requires a decision-sufficient observation:  $o_t$  must expose the facts the exact teacher uses to resolve step  $t$ , such as the grounded successor candidates and visited membership, so that a correct operation is reachable from the observation alone.

The policy output at step  $t$  is a single Typed Search Operation  $u_t = (\tau_t, \theta_t)$ . The operation type  $\tau_t$  comes from the vocabulary of the declared algorithm, covering steps such as expanding a node, generating a successor, updating the frontier, or applying a goal test. The operand vector  $\theta_t$  carries the arguments of that step. Ownership is defined over operands. Every operand that determines exploration must be emitted by the policy, including the selected source or frontier state and any successor, action, or insertion decision the declared operation requires. An emission that omits an exploration-determining operand, or names operands inconsistent with the decision context, is invalid by definition.

Search Memory  $M_t$  is the external state the Trusted Search Runtime maintains across an episode. It holds the frontier, the visited or best-depth table, and algorithm-specific state such as novelty counters for BFWS. It is bounded, declared per protocol cell, and it is not internal unbounded model state. The policy has no access to episode history except through what the adapter exposes from  $M_t$  in  $o_t$ . Equivalently, the observation sequence is the only channel through which prior decisions influence later ones.

Each declared algorithm carries Algorithm Invariants, deterministic properties the runtime checks on every operation. For BFS, the frontier must be consumed in first-in-first-out order. For BFWS, selection must follow novelty and goal-count priority with the declared duplicate handling. For the additive best-first variants, frontier order must follow  $f(s) = g(s) + w \cdot h_{\text{add}}(s)$  with  $w = 3$ , where  $g(s)$  is the accumulated cost of the path that reached  $s$ , or  $h_{\text{add}}$  alone for greedy best-first, under the declared tie-breaking rule. Stepwise validity and invariant compliance are execution-quality measurements. They do not establish termination, completeness, optimality, or episode success, which are reported separately under each cell’s declared task, heuristic, tie-breaking rule, and budget.

The Trusted Search Runtime has four permitted actions on an emission: validate it, apply it, persist its effects to Search Memory, or reject it. It may not choose omitted operands, reorder candidates on the policy’s behalf, silently repair an invalid emission, or substitute a default search decision. This boundary gives the attribution test used throughout the paper. Replaying the raw policy emissions  $u_{1:T}$  against the same task and the same initial Search Memory must reproduce the explored trace exactly. A decision that appears only because the runtime filled, reordered, or repaired an emission is runtime behavior, not model behavior, and the replay test detects it.

A single step under BFS illustrates the contract. The observation lists the frontier in order and, for the current source state, the grounded successor candidates with their visited membership. To expand, the policy emits an operation whose operands name the source state and, for each insertion, the chosen successor and its enqueue position. The runtime validates that the named source is the frontier head, that each named successor is an unvisited grounded successor of that source, and that each insertion lands at the tail, then applies and persists the update. Any failed check rejects the emission. Nothing is repaired, and no alternative source, successor, or position is chosen on the policy’s behalf.

The training unit is a Search-Trace Segment, a bounded ordered slice of Typed Search Operations and the observations that preceded them, cut from replay-verified traces of exact algorithm execution. Process supervision is behavior cloning over a dataset  $\mathcal{D}$  of such segments. The policy is trained with token-level cross-entropy to emit the teacher’s operation given the observation, so supervision lands on the exploration decisions themselves rather than on a final plan. An optional DAGger stage (Ross et al., 2011) rolls the current policy out under the runtime, collects teacher corrections on the decision contexts the policy actually visits, aggregates them into  $\mathcal{D}$ , and retrain. Both conditions train against the same contract above. The arms, exposure matching, and evaluation gates that compare them are defined in Section 4.

## 4 EXPERIMENTAL DESIGN

The study is organized around one comparative question per cell, asked of every algorithm-by-modality-by-condition combination: whether a trained Search Process Policy exceeds an oracle-assisted validity control on full-episode budgeted success under the ownership contract of Section 3. Training and evaluation run under frozen protocols and a 336 GPU-hour program cap, so the work proceeds as a staged sequence in which each stage is admitted by a cost gate and closed by a pre-specified outcome (PASS, VALID\_STOP, INVALID, or ANCESTOR\_STOP) that is preserved as a receipt. Every executed stage is development-stage evidence. Panels are frozen and outcome-blind, the held-out manifest has never been accessed, and all model comparisons use a single training seed per cell except the two cells replicated under issue #127. We report the design with its gate statuses intact, including the stages that stopped early, because the stop receipts bound what the evidence supports.

Four declared algorithms form the active matrix. Breadth-first search (BFS) provides the minimal frontier discipline. The structural gate uses the complete unpruned variant of best-first width search (Lipovetzky & Geffner, 2017), which orders the frontier by novelty and goal count without pruning. Two additive best-first variants complete the matrix, weighted A\* with weight 3 (Pohl, 1970) and greedy best-first, both over the additive heuristic  $h_{\text{add}}$  (Bonet & Geffner, 2001) and named `astar_w3` and `astar_greedy` in the transfer branch. An earlier stage paired two optimal A\* variants that differed in heuristic representation, one over  $h_{\text{max}}$  and one over a landmark-count heuristic. That pair is retired. A governed VALID\_STOP under issue #62 ended the stage before corpus construction, its heuristic-representation estimand remains unidentified, and no optimal A\* arm participates in the active design.

Each algorithm-by-modality cell holds three checkpoint conditions. The base condition is the pre-trained model with no task-specific optimization. In the expanded program, the process-SFT condition trains a fresh LoRA adapter (Hu et al., 2022) on replay-verified Search-Trace Segments, using 512 unique records for one epoch (16 optimizer updates) per cell. Targets are the exact teacher’s Typed Search Operations. The BFWS structural-gate checkpoint predates this recipe. It trained on the full 47,780-record training corpus for two epochs (seed 17). The additive development panels (issues #65 and #66) likewise used two-epoch full-corpus runs. The third condition continues a process-SFT checkpoint with two iterations of DAgger (Ross et al., 2011), in which the exact teacher relabels the states that the learned policy actually visits. An exposure-matched continued-SFT arm receives the same added training exposure and controls for it. The primary backbone is Qwen3-VL-8B-Instruct (Bai et al., 2025) at pinned revision `0c351dd0`, and the second-backbone stage trains InternVL3.5-8B (Wang et al., 2025) under the same expanded-program cell recipe.

Exactly one training seed is authorized per model/cell configuration under the supervisor budget, and every executed cell after the pilot uses seed 17, except the two cells selected for prospective training-seed replication (issue #127, protocol frozen 2026-09-20) that added seeds 29 and 71. The historical BFS pilot predates this policy. It trained five process-SFT seeds on the bounded v8 corpus and evaluated all five on its development panel. That five-seed run is retained as pilot history and is not a template for later cells. Outside the pilot and the two replication cells, no cell estimates training-seed variance, and repeated rollout or reference seeds do not substitute for a training seed.

Two programmatic references bracket every learned arm. The exact-classical reference is the declared algorithm itself, executed by the Trusted Search Runtime with replay verification under the same budget, and it bounds perfect operation-level decision-making. The random-valid reference is an oracle-assisted control that proposes only operations the runtime will accept as valid, so it bounds what valid-operation bookkeeping alone achieves without learned state evaluation. Neither reference is a learned ability. When a reference saturates a panel, the learned-to-control gain on that panel is unmeasurable rather than zero. [TODO: random-valid definition] The frozen protocol has not yet pinned the validity-constrained sampling distribution, the candidate access granted to the control, the runtime filtering step, or the explicit absence of retry and repair. These details are owed to the evidence contract before the final evaluation can freeze.

All splits are by whole task instance, so no state, decision, or target from an evaluation task appears in any training row, and each released corpus passed no-overlap, no-conflict, live-parity, and token-budget audits before use. The panel inventory is frozen. The BFS pilot used an outcome-blind 15-task development panel. Its released text-state process corpus holds 25,109 process rows (12,994

train, 12,115 dev) drawn from 90 replay-verified traces. The BFWS gate froze a 105-task development panel and a 45-task held-out manifest that remains unaccessed, and its 69,019 replay-verified teacher decisions project to 47,780 train and 21,239 dev process rows. The modality matrix evaluates 24 tasks from 12 domains under three Modality Observations (text-state, visual-state, multimodal-state), which gives 288 episodes per arm across the four algorithms and 1,152 bindings across the four arms. The generalization stage admits 25 frozen semantics-preserving perturbation variants, yielding 150 learned-adaptor episodes inside 1,200 replayed episodes. The observation-corruption suite freezes  $k = 9$  source tasks. The transfer stage runs 36 paired comparisons.

Episodes are budgeted in decision calls, and invalid operations are charged to the budget. Budget parity holds inside each cell. Learned adapters are capped at twice the exact-reference decision count of the task, the base condition is priced at one call per episode, and the programmatic references run on CPU under the same frozen rollout rules. Under those rules an episode terminates on its first invalid emission, so model arms have no retry or repair path. A model episode counts only after an independent replay of its raw emissions reproduces the recorded trace. Full-episode budgeted success is the primary measure. Operation validity, Algorithm Invariant compliance, invalid-operation rate, and charged budget are reported separately, and no stage combines them into one score.

Contrasts are paired by whole task instance. Unless a stage’s frozen protocol states otherwise, paired differences use a bootstrap over instances with 10,000 resamples at seed 1729 and 95% percentile intervals, transfer comparisons use exact McNemar tests against the base with Holm correction across the 36 comparisons, and strata with fewer than 8 instances are reported as descriptive only. Under one training seed per cell (three in the two replicated cells) and panels of this size, these intervals are descriptive bounds and cannot support broad superiority claims.

The staged sequence, with the status of each stage at the evidence snapshot of 2026-09-21, is as follows.

1. **Bounded BFS pilot (executed, closed as history).** Five process-SFT seeds on the bounded v8 corpus reached invariant-valid success on the frozen 15-task outcome-blind panel, but random-valid matched them, so the stage closed with a governed VALID.STOP and downstream work was gated-not-run. Capped-iteration width search survives from this era as qualification history and plays no role in the active gate.
2. **BFWS structural gate (executed, development-panel PASS).** The complete unpruned BFWS successor replaced capped iteration width after independent IW(1), IW(2), and IW(3) attempts proved not corpus-complete. The 105-task panel, the replay-verified traces, and the released process corpus qualify this stage, and a frozen 15-task text-state development panel under BFWS supplies what Section 5.2 identifies as the study’s only positive learned-control contrast to date. The BFWS gate evaluated a preregistered exact-cost fallback panel. It selected the cheapest exact-decision task in each of the 15 domains from the 35-task development pool before any model outcome, breaking ties by difficulty and instance ID. The BFS pilot’s 15-task v8 panel instead came from the v6 development set, with one task per domain (11 easy, 3 medium, 1 hard). [TODO: evidence] The runbooks do not pin the task overlap between these panels or its count.
3. **Modality matrix (executed).** The expanded matched baseline crosses the 24 tasks, 12 domains, four algorithms, and three modalities described above. Two ablations sit beside it. Curriculum-by-modality trains nine cells (three modalities under staged, shuffled, and mixed orderings of greedy best-first records) and evaluates 27 unseen tasks. The successor-prediction stage replaces trusted successors with model-generated ones over 90 episodes in 15 modality-by-panel cells on a 12-task fallback panel, separating end-to-end competence from structural control under trusted successors.
4. **DAGger (executed).** Two iterations on BFS, evaluated against the exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs per modality (72 unseen rows).
5. **Generalization and robustness v2 (executed at reduced frozen scope).** The frozen admission selected 25 of 93 eligible variants as a uniform cheapest-prefix sample of five per family. [TODO: full-scope v1] The full v1 suites stopped at their cost-admission gates and remain unexecuted (issues #96 and #98 open).
6. **Second backbone v2 (executed at reduced frozen scope).** InternVL3.5-8B (pinned OpenGVLab/InternVL3.5-8B-HF revision 741a7d03) was trained on a 12-task key-cell

panel selected by lowest reference BFS decision count and frozen before any outcome. Because this backbone shares the Qwen3-8B language-model family with the primary model, the contrast isolates the vision tower, connector, image tokenization, and multimodal recipe rather than the language model. The full-scope v1 branch remains unexecuted (issues #102 and #103 open).

7. **Failure-mechanism calibration (executed, CPU-only).** A frozen failure taxonomy was applied to 2,910 replay-verified episodes, classifying 905 failures with zero unclassified, to locate where learned execution fails relative to the base and the oracle-assisted control.
8. **Observation-corruption suite (executed).** Frozen checkpoints, with no retraining, were evaluated under four deterministic corruption families applied at observation time (visual blanking, visual degradation, seeded text shuffling, and text masking) on the  $k = 9$  frozen tasks, probing which Modality Observation channel carries the learned control.
9. **Zero-shot transfer (executed).** All 12 verified planning adapters plus the base were evaluated on frozen subsets of FOLIO (200 items), GSM8K (200 items), and HumanEval (164 items) under the 36-comparison paired contract described above.
10. **Final primary evaluation (pending).** [TODO: final evaluation] No authorized held-out evaluation exists. Freezing one requires recorded decisions on the following prerequisites.
  - The condition-level data source and Search-Trace Segment construction with per-algorithm allocation.
  - Compute parity and checkpoint-selection rules.
  - The episode-budget unit, the whole-instance split procedure, and the fixed Search Memory capacity.
  - The estimand, the uncertainty interval or test, the success threshold, the invariant-compliance rule, and the validity and invalid-operation denominators for every cell.
 Until these choices are set, the manuscript is a result-ready empirical skeleton rather than a pre-registered evaluation protocol.
11. **Training-seed replication (executed).** A prospective amendment (protocol expanded-seed-replication-v1, issue #127, frozen 2026-09-20 before any seed artifact existed) added training seeds 29 and 71 to two cells selected by decision relevance, not by outcome: cell A, the headline process-SFT cell (`best_first_add_greedy` × multimodal-state, 24-problem unseen panel), and cell set B, DAGger iteration-1 versus exposure-matched continued-SFT across three modalities (72 unseen rows per seed). Seeds, cell selection, evaluation contracts, and analysis were frozen before launch. No outcome-based cell swapping occurred and no seed was dropped after outcomes. All 534 new episodes (48 baseline, 486 DAGger) were independently replayed under the original frozen evaluation contracts. Cross-seed aggregation is descriptive ( $n = 3$ ), and all cells outside these two remain single-seed (17).

One further arm is specified but unmeasured: image-only state-history observation conditions (visual-nomem-state and visual-seq-state, issue #128) exist as a frozen-protocol proposal without GPU authorization under the current program. [TODO: image-only arms] Measurements for every executed stage, including the nulls and the stop receipts, are reported in Section 5.

## 5 RESULTS

This section reports the executed evidence in gate order. Gate receipts and frozen scopes come first. Effect sizes are interpreted only inside the scope their gates passed. Throughout, full-episode budgeted success, operation validity, Algorithm Invariant compliance, invalid-operation charges, budget usage, and learned-to-control gain are reported as separate measurements and are never combined into a single score. The random-valid and exact-classical references are oracle-assisted bounds, not learned abilities, and where they saturate they bound what the comparison can measure rather than certify policy quality. Except for two prospective replication cells (Section 5.6), every learned arm rests on a single training seed (17). Every panel below is a development panel. The held-out final primary evaluation has not been run, so no finding here is a final comparative efficacy claim.

### 5.1 GATE RECEIPTS AND EXECUTED SCOPE

The expanded study program ran on 2 A100 80GB GPUs under a 336 GPU-hour cap with an absolute scheduler cutoff, and spent 53.52 of those hours (15.9%) at the #120 synthesis snapshot, before

Table 2: Gate receipts for the expanded study program, before effect interpretation. Terminal states and frozen scopes are copied from the branch reconciliation ledger. The two reduced-scope branches closed on v2 evidence. Their v1 scopes stay open as VALID\_STOP history.

| Branch                    | Tickets         | Terminal state             | Frozen scope executed                    | Coverage             |
|---------------------------|-----------------|----------------------------|------------------------------------------|----------------------|
| Expanded baseline         | #116–#118       | complete                   | 1,152 bindings, 576 model episodes       | 1,152/1,152 replayed |
| DAGger                    | #78–#84         | complete                   | 405 episodes + 81 paired rows            | 405/405              |
| Successor prediction      | #85–#89, #99    | complete                   | 90 episodes, 15 modality-panel cells     | 90/90                |
| Curriculum modality       | by #119         | complete                   | 243 model + 324 comparator bindings      | all present          |
| Transfer                  | #104–#107       | complete                   | 39 cells: 12 v1 + 27 v2                  | 39/39                |
| Second backbone           | #101–#103, #123 | v1 VALID_STOP. v2 complete | 72 model episodes + 72 comparators       | 72/72                |
| Generalization/robustness | #96, #98, #124  | v1 VALID_STOP. v2 complete | 1,200 episodes: 300 model + 900 controls | 1,200/1,200          |
| Observation corruption    | #126            | complete                   | 360 new bindings                         | 360/360 replayed     |
| Training-seed replication | #127            | complete                   | 534 new episodes                         | 534/534 replayed     |

the #126/#127 addenda. Seven branches were admitted. Five executed to completion at v1 scope, and two (second backbone, generalization/robustness) stopped with terminal VALID\_STOP receipts at their frozen v1 cost-admission gates and later completed at reduced scope under new versioned protocols. Table 2 lists the per-branch receipts with their frozen scopes. Every executed branch reconciles with zero missing evidence: 1,152/1,152 baseline episodes independently replayed, 405/405 DAGger episodes, 90/90 successor episodes, 243/243 curriculum model episodes with 324/324 comparator bindings, 39/39 transfer cells, 72/72 second-backbone-v2 model episodes with 72 hash-pinned comparators, and 1,200/1,200 generalization-v2 episodes. The v1 full scopes of the two reduced branches were never executed. Their terminal VALID\_STOP evidence is retained as history, and the corresponding issues remain open [TODO: full-scope v1 branches #96/#98/#102/#103].

Three earlier gates frame the active algorithm matrix. The bounded BFS pilot closed with a governed VALID\_STOP: on a frozen outcome-blind 15-task BFS development panel, final checkpoints from five process-SFT seeds achieved 1.0 invariant-valid success with a 0.0 invalid-operation rate, but the base achieved 0.0 and random-valid achieved 1.0, so the learned policy showed executable BFS behavior with zero gain over the strongest control and downstream work was gated off. The BFWS structural gate then passed on its frozen development panel (Section 5.2). The companion 45-task held-out manifest was frozen before outcomes and has never been accessed [TODO: held-out final primary evaluation]. The optimal A\* arm over  $h_{\max}$  with landmark counting was retired after its own governed VALID\_STOP and is superseded by the two additive best-first variants. Its original heuristic-representation estimand remains unidentified. The historical five-seed BFS pilot predates the current budget rule, which authorizes one training run at seed 17 per model and cell (Section 4).

## 5.2 PRIMARY MATRIX: SUCCESS BY ALGORITHM AND CONDITION

The single positive learned-control comparison (confirmed by the expanded baseline below) comes from the frozen 15-task text-state BFWS development panel (Table 3). One fixed-seed process-SFT run achieved 0.933 success, versus 0.467 for the oracle-assisted random-valid control, 0.0 for the pretrained base, and 1.0 for exact execution. The absolute gain over random-valid is 0.467, with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). This is a development-stage result from one training seed, and the held-out manifest was not accessed. On the BFS and the two additive best-first development panels, process SFT matched random-valid at 100%. Those controls saturate, so executable behavior on these panels does not identify learned advantage. Saturation removes measurability rather than demonstrating ability.

The expanded matched baseline tightens this picture. It covers 24 whole problems from 12 domains under all four active algorithms and all three modality adapters: 24 problems  $\times$  4 algorithms  $\times$  3 modalities  $\times$  4 arms gives 1,152 logical bindings and 576 model episodes, all independently

Table 3: Frozen 15-task text-state BFWS development panel: full-episode budgeted success by condition. The learned-to-best-control gain over random-valid is 0.467 with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). Single fixed training seed, development stage, held-out manifest not accessed.

| Condition                              | Success (15 tasks) |
|----------------------------------------|--------------------|
| Pretrained base                        | 0.0                |
| Process SFT                            | 0.933              |
| Random-valid (oracle-assisted control) | 0.467              |
| Exact execution                        | 1.0                |

Table 4: Primary matrix: expanded 24-task baseline by algorithm and modality. Measures are reported separately and never combined. Columns give full-episode budgeted success out of 24 for each condition, process-SFT invalid operations over decisions (inv/dec), process-SFT expansions, and exact-reference decisions as the budget reference. The pretrained base terminates on its first invalid operation in every cell (24/24 inv/dec, 0 expansions). Random-valid (RV) is an oracle-assisted programmatic control and exact execution is the oracle upper bound, and both are references rather than model abilities.

| Algorithm             | Modality   | Base | SFT   | RV    | Exact | SFT inv/dec | SFT exp. | Exact dec. |
|-----------------------|------------|------|-------|-------|-------|-------------|----------|------------|
| BFS                   | text       | 0/24 | 0/24  | 15/24 | 24/24 | 24/66       | 15       | 1969       |
| BFS                   | visual     | 0/24 | 0/24  | 15/24 | 24/24 | 24/69       | 17       | 1969       |
| BFS                   | multimodal | 0/24 | 0/24  | 15/24 | 24/24 | 24/63       | 15       | 1969       |
| BFWS                  | text       | 0/24 | 0/24  | 17/24 | 24/24 | 24/24       | 0        | 974        |
| BFWS                  | visual     | 0/24 | 0/24  | 17/24 | 24/24 | 24/24       | 0        | 974        |
| BFWS                  | multimodal | 0/24 | 0/24  | 17/24 | 24/24 | 24/24       | 0        | 974        |
| Weighted A* ( $w=3$ ) | text       | 0/24 | 20/24 | 24/24 | 24/24 | 4/609       | 179      | 694        |
| Weighted A* ( $w=3$ ) | visual     | 0/24 | 18/24 | 24/24 | 24/24 | 6/499       | 143      | 694        |
| Weighted A* ( $w=3$ ) | multimodal | 0/24 | 22/24 | 24/24 | 24/24 | 2/666       | 196      | 694        |
| Greedy best-first     | text       | 0/24 | 22/24 | 24/24 | 24/24 | 2/614       | 179      | 673        |
| Greedy best-first     | visual     | 0/24 | 22/24 | 24/24 | 24/24 | 2/643       | 189      | 673        |
| Greedy best-first     | multimodal | 0/24 | 21/24 | 24/24 | 24/24 | 3/617       | 179      | 673        |

replayed. Table 4 gives the per-cell matrix. In aggregate, the pretrained base succeeds on 0/288 instances and every one of its episodes terminates on an invalid operation at the first decision. Process SFT succeeds on 125/288 with 163 invalid operations charged across its episodes. Random-valid succeeds on 240/288 and exact execution on 288/288. Learned success is confined to the additive best-first family: with BFS or width-based best-first bookkeeping, process SFT fails 0/24 in every modality, always terminating on an invalid operation.

Paired whole-problem contrasts supply the learned-to-best-control gain, kept separate from the success counts (Table 5). Against the base, process SFT gains only in the additive cells (+0.833 for weighted A\*, +0.903 for greedy) and gains nothing under BFS or BFWS. Against random-valid, process SFT loses in every algorithm column and wins zero of the 72 problem-modality units in any column. Random-valid itself ties exact execution in both additive columns, so those cells cannot measure any advantage beyond valid bookkeeping. These contrasts are tiny-subgroup descriptive bounds over 24 problems and one training seed. They do not support modality or algorithm superiority claims beyond the executed panel. The development-panel BFWS positive does not reappear on this suite. The complete six-measure matrix for the frozen final protocol, including per-cell Algorithm Invariant compliance and charged-budget columns, belongs to the held-out evaluation [TODO: held-out final primary evaluation].

### 5.3 SECONDARY PANELS

**Modality.** The per-modality rows of Table 4 show the same qualitative profile in all three adapters: zero learned success under BFS and BFWS, learned success below the saturating controls in the additive cells. The earlier matched-modality study, a 3-problem panel on its own ledger that is never pooled with the expanded baseline, gives process-SFT success of 5/12 in text-state, 6/12 in visual-state, and 6/12 in multimodal-state, versus 10/12 for random-valid in each modality, 0/12 for the base, and 12/12 for exact execution. The evidence does not support an intrinsic modality

Table 5: Paired whole-problem success contrasts on the expanded baseline (10,000 paired resamples over the 24 problems, seed 1729). Entries give the point contrast with wins/losses/ties over the 72 problem-modality units. Contrasts are descriptive bounds over a tiny subgroup and a single training seed (Efron, 1979).

| Contrast             | BFS              | BFWS             | Weighted A* ( $w=3$ ) | Greedy          |
|----------------------|------------------|------------------|-----------------------|-----------------|
| SFT – base           | +0.000 (0/0/72)  | +0.000 (0/0/72)  | +0.833 (60/0/12)      | +0.903 (65/0/7) |
| SFT – random-valid   | −0.625 (0/45/27) | −0.708 (0/51/21) | −0.167 (0/12/60)      | −0.097 (0/7/65) |
| SFT – exact          | −1.000 (0/72/0)  | −1.000 (0/72/0)  | −0.167 (0/12/60)      | −0.097 (0/7/65) |
| Random-valid – exact | −0.375 (0/27/45) | −0.292 (0/21/51) | +0.000 (0/0/72)       | +0.000 (0/0/72) |

Table 6: DAgger on BFS: development and unseen success with unseen invalid-operation rates, against the exposure-matched continued-SFT control and the two references.

| Arm                       | Dev success /9 | Unseen success /72 | Unseen invalid rate |
|---------------------------|----------------|--------------------|---------------------|
| DAgger iteration 2        | 3/9            | 1/72               | 0.316               |
| Continued SFT iteration 2 | 2/9            | 1/72               | 0.175               |
| Original process SFT      | 1/9            | 0/72               | 0.364               |
| Random-valid              | 6/9            | 45/72              | 0.000               |
| Exact reference           | 9/9            | 72/72              | 0.000               |

effect. Two caveats bound any such claim: the unlabelled 128px rendered views can lose symbolic identities, and the static and goal pages still contain text in every modality, so semantic parity across adapters is not established.

**Generalization and robustness v2.** The reduced v2 protocol admitted a uniform cheapest-prefix sample of 5 variants per family over 5 perturbation families (25 of 93 eligible semantics-preserving variants, membership frozen before execution) and evaluated the learned additive-best-first adapters. Learned adapters succeeded on 128/150 episodes versus 0/150 for the pretrained base, where every base episode is a one-call invalid termination (150 decisions total). By family, learned success was 29/30 on object-renaming, 27/30 on name-compression, 27/30 on render-restyle, 26/30 on scale-up, and 19/30 on shifted-init. By modality the split was text 44/50, multimodal 43/50, visual 41/50; by algorithm, greedy 67/75 and weighted A\* 61/75. All 22 learned failures are single-invalid-operation terminations, 11 of them in shifted-init. Controls saturate (random-valid 750/750, exact 150/150) and remain bounds, not learned ability. The lossy name-compression stratum is reported separately and never pooled: all 5 admitted variants are text-lossy, 3 multimodal-lossy, and 0 visual-lossy, with learned success on that stratum of text 9/10, visual 8/10, multimodal 10/10. Family strata hold 5 problems each, so family-level rates are descriptive only. The v1 full suite remains unexecuted [TODO: full-scope v1 generalization branches #96/#98].

**DAgger.** Two DAgger iterations (Ross et al., 2011) on BFS were compared against an exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs (Table 6). On the unseen panel, DAgger and continued SFT each achieve 1/72 invariant-valid successes (paired over whole problems: one DAgger win, one loss, 70 ties), both far below random-valid at 45/72 and exact execution at 72/72, while the original process-SFT checkpoint achieves 0/72. The result is null. DAgger also carries the higher unseen invalid-operation rate of the two iterated arms (0.316 versus 0.175). Final cumulative unique training corrections were text 201, visual 203, multimodal 181, accumulated over 12 training cells of 512 records and 16 optimizer updates per cell-iteration.

**Model-generated successors.** Replacing trusted successors with model-generated dynamic-state successors collapses goal success from 45/45 to 1/45 on the evaluated panel, with zero trusted-state substitutions in the trusted arm. The single success is the one multimodal unseen cell. The trained successor model is exactly correct on 65/109 retained raw predictions, with 93 schema-valid and 66 effect-valid, and downstream failure kinds are dominated by exact state-identity and effect failures (27 effect, 16 schema, 1 static-context) rather than schema failures. Collection-time verification of the untrained exact-successor accepted 119/1,536 predictions (text 43, visual 52, multimodal 24). Coverage is partial by design: the frozen outcome-blind admission accepted a 12-task fallback panel after the full 24-task unseen panel was priced at 147.66 GPU-hours against the 64 GPU-hour branch cap, so full unseen coverage remains missing.

Table 7: Zero-shot transfer accuracy on frozen benchmark subsets (FOLIO 200, GSM8K 200, HumanEval 164). Adapter rows follow the transfer ledger names: `bfs`, `iw` (width-based best-first, BFS), `astar_w3`, and `astar_greedy`, each in text (t), visual (v), and multimodal (m) training cells. No comparison against the base survives Holm correction (0/36; minimum adjusted  $p$  0.8156).

| Adapter                         | FOLIO                 | GSM8K                 | HumanEval             |
|---------------------------------|-----------------------|-----------------------|-----------------------|
| Base                            | 0.600                 | 0.955                 | 0.780                 |
| <code>bfs t/v/m</code>          | 0.600 / 0.605 / 0.605 | 0.950 / 0.945 / 0.965 | 0.793 / 0.799 / 0.780 |
| <code>iw t/v/m</code>           | 0.660 / 0.645 / 0.645 | 0.960 / 0.965 / 0.960 | 0.768 / 0.762 / 0.756 |
| <code>astar_w3 t/v/m</code>     | 0.590 / 0.575 / 0.590 | 0.950 / 0.965 / 0.945 | 0.817 / 0.799 / 0.774 |
| <code>astar_greedy t/v/m</code> | 0.590 / 0.590 / 0.590 | 0.950 / 0.955 / 0.950 | 0.799 / 0.811 / 0.793 |

**Curriculum by modality.** Nine training cells (three modalities under staged, shuffled, and mixed ordering, all under greedy additive best-first and trained fresh from the same base on identical record sets) were evaluated on 27 unseen whole problems. Successes were 24/24/25 (staged/shuffled/mixed) in text-state, 24/27/26 in visual-state, and 25/27/27 in multimodal-state. A sequential-SFT comparator reached 24/27, 25/27, and 24/27 in text, visual, and multimodal cells respectively, the base stayed at 0/27, and random-valid and exact execution saturated at 27/27 in every modality. All four paired interaction intervals include zero:  $-0.111 [-0.259, +0.037]$  and  $-0.074 [-0.259, +0.074]$  for the visual and multimodal staged-minus-shuffled contrasts against text, and  $-0.074 [-0.259, +0.111]$  and  $-0.037 [-0.185, +0.111]$  for the corresponding mixed-minus-shuffled contrasts. No material ordering-by-modality interaction emerged. Because both references saturate and the base sits at floor, ceiling and floor effects bound the observable differences.

**Transfer.** All 12 verified planning adapters plus the base were evaluated zero-shot on frozen subsets of FOLIO (Han et al., 2022) (200 items), GSM8K (Cobbe et al., 2021) (200), and HumanEval (Chen et al., 2021) (164). Table 7 gives the per-adapter accuracies. The paired verdict uses McNemar exact comparisons (McNemar, 1947) of each adapter against the per-example base predictions, Holm-corrected across all 36 comparisons (Holm, 1979). None survives correction: the minimum adjusted  $p$  is 0.8156, and the smallest raw  $p$  is 0.0227 (FOLIO with the text width-based adapter, at +12 examples). Leakage screening over the frozen 8-gram window finds 7/564 benchmark items sharing only trivial numeric 8-grams with the planning corpus and zero content overlap. The transfer result is a null.

**Second backbone.** The reduced v2 replication trained and evaluated InternVL3.5-8B (Wang et al., 2025) on a frozen 12-task key-cell panel (selected by lowest reference BFS decision count before any model outcome), one seed-17 training run per modality cell. Neither arm reaches a goal: the pretrained base scores 0/36 running 1 decision per episode (36 total), and process SFT scores 0/36 running 1 to 4 decisions (59 total). Reused controls score random-valid 54/72 (18/24 per modality, oracle-assisted) and exact 72/72. Paired contrasts give SFT minus base  $+0.000 [+0.000, +0.000]$  and SFT minus random-valid  $-0.750 [-1.000, -0.500]$  in all three modalities. The cross-backbone SFT contrast against the primary Qwen3-VL runs is  $+0.000 [+0.000, +0.000]$  in every modality, but it is descriptive rather than paired, because the pinned baseline evidence holds aggregate cells only. InternVL3.5-8B shares the Qwen3-8B LLM family (Yang et al., 2025), so this contrast isolates the vision tower, connector, image tokenization, and multimodal recipe, not the LLM backbone. The v1 full-panel scope remains unexecuted [TODO: full-scope v1 second-backbone branches #102/#103].

#### 5.4 FAILURE-MECHANISM CALIBRATION

A frozen CPU-only calibration mined 2,910 unique replay-verified episodes across the five executed model branches and classified all 905 failures with zero unclassified. The protocol was frozen before any outcomes were computed, and the analysis is descriptive. Three arm-level profiles emerge. The base policy fails in 447 of 447 episodes, always on the first decision: 379 emit a malformed output that never reaches the operation checker, and 67 issue rejected BFS frontier operations. Learned arms fail in 407 of 906 episodes, and the dominant learned failure kind is `other_invariant_violation` with 229 occurrences (56.3% of learned failures): parsed, source-correct, applicable operations that the runtime rejects because they violate the declared search

Table 8: Observation corruption on frozen checkpoints ( $k=9$  admitted tasks, paired whole-problem deltas against clean runs, strata below 8 descriptive-only under the frozen rule). “Material” is the frozen protocol’s interval-based verdict.

| Corruption      | Arm        | Corrupted | Clean | $\Delta$ | 95% interval       | Material |
|-----------------|------------|-----------|-------|----------|--------------------|----------|
| Text-masked     | text       | 0.000     | 0.944 | -0.944   | $[-1.000, -0.833]$ | yes      |
| Text-masked     | multimodal | 0.000     | 0.889 | -0.889   | $[-1.000, -0.722]$ | yes      |
| Text-shuffled   | text       | 0.000     | 0.944 | -0.944   | $[-1.000, -0.833]$ | yes      |
| Text-shuffled   | multimodal | 0.167     | 0.889 | -0.722   | $[-0.889, -0.500]$ | yes      |
| Visual-blank    | visual     | 0.833     | 0.889 | -0.056   | $[-0.167, +0.000]$ | no       |
| Visual-blank    | multimodal | 0.778     | 0.889 | -0.111   | $[-0.278, +0.000]$ | no       |
| Visual-degraded | visual     | 0.833     | 0.889 | -0.056   | $[-0.167, +0.000]$ | no       |
| Visual-degraded | multimodal | 0.778     | 0.889 | -0.111   | $[-0.278, +0.000]$ | no       |

invariant. The remaining learned failures are 72 malformed outputs, 56 inapplicable grounded actions, 27 effect failures, 16 schema failures, and smaller kinds. First failures cluster early, at decision indices 0, 1, and 2 with counts 137, 88, and 46. The oracle control fails 51 times, all late and all by expansion-budget exhaustion (buckets 5–8, 9–16, 17–32, and 33+ hold 9, 9, 3, and 30).

Every baseline process-SFT episode under BFS or BFWS fails (144/144). The DAgger-iteration BFS arms fail 155/162, while the learned additive best-first families fail 56 of 546 episodes (10.3%). This pattern matches the primary matrix (Section 5.2). Read together with the horizon profile, process supervision teaches the policy to emit well-formed, source-correct operations, but not to maintain invariant-consistent frontier intent over a full episode: the learned policy fails early on invariant grounds, whereas the oracle control, which always emits valid operations, fails only when its expansion budget runs out late in the episode.

## 5.5 OBSERVATION-CORRUPTION CHANNEL TESTS

The observation-corruption suite evaluated frozen checkpoints, with no retraining, on a prospectively frozen set of  $k=9$  admitted tasks under four deterministic corruption families applied at observation time. Clean learned success on these tasks is 0.917 (text-state 0.944, visual-state 0.889, multimodal-state 0.889; 9 tasks  $\times$  2 algorithms per cell). Table 8 gives the paired whole-problem deltas. Language-channel corruption is catastrophic: masking or shuffling the text channel drops success by up to  $-0.944$   $[-1.000, -0.833]$ , and the multimodal arm retains at most 0.167 success under shuffled text. Visual-channel corruption is not material: blanking or degrading the rendered state costs at most  $-0.111$   $[-0.278, +0.000]$ , and every visual interval includes zero. The multimodal channel-isolation contrast, paired by task and algorithm ( $n=18$ ), is  $-0.806$  for text-channel corruption versus  $-0.111$  for visual-channel corruption, giving text minus visual  $-0.694$  with 95% interval  $[-0.889, -0.472]$ , which is material under the frozen rule. Controls behave as the contract predicts: the corrupted base stays at floor (0/144, all first-decision invalid), random-valid succeeds 72/72 on the new frozen seeds, so per the saturation rule no structural-advantage claim is made, and exact execution remains at 18/18.

Under the frozen observation contract, the learned policy depends on the language observation. Corrupting that channel is catastrophic even when clean rendered images remain available, while removing or degrading the visual channel barely moves success, including on the visual-state arm whose only rendered-semantics channel was corrupted. The shared textual search-memory and candidate serialization carries the decision-relevant state. This is a channel finding about the trained controller under the declared adapters, not evidence about intrinsic modality value: rendered visual semantics are near-dispensable under these adapters because the textual scaffold already exposes them.

## 5.6 DEVELOPMENT-STAGE BOUNDARY

All results above are development-stage. Every learned cell uses one training run at seed 17 except the two prospective replication cells below. Three evaluations remain pending and bound every claim in this paper. The held-out final primary evaluation on the frozen 45-task manifest has not been executed, so no final comparative efficacy finding exists [TODO: held-out final primary evaluation]. The prospective training-seed replication (issue #127, seeds 17/29/71 frozen pre-launch, 534 new

Table 9: Prospective training-seed replication for the headline process-SFT cell (greedy best-first  $\times$  multimodal-state, 24-problem unseen panel). Seeds 17/29/71 frozen pre-launch; 534 new episodes independently replayed; aggregation descriptive ( $n=3$ ).

| Measure            | Seed 17 | Seed 29 | Seed 71 |
|--------------------|---------|---------|---------|
| Success /24        | 21/24   | 21/24   | 22/24   |
| SFT – base         | +0.875  | +0.875  | +0.917  |
| SFT – random-valid | −0.125  | −0.125  | −0.083  |

episodes independently replayed) executed on two cells (Table 9). On the headline process-SFT cell (greedy best-first  $\times$  multimodal-state, 24-problem unseen panel), per-seed success is 21/24, 21/24, 22/24 (mean 0.889, range [0.875, 0.917]). The contrast against the pretrained base is positive at every seed (+0.875/+0.875/+0.917, no sign flip). The contrast against random-valid stays negative (−0.125/−0.125/−0.083). On the DAGger iteration-1 cell set (BFS  $\times$  three modalities, DAGger and continued-SFT arms), every arm and seed scores 0/72 unseen, with paired contrast +0.000 [+0.000, +0.000] at each seed. The published iteration-2 verdict (1/72 vs 1/72, seed 17) is unchanged. Both cells replicate across all three seeds. Cell A covers one of the three pooled headline modalities. The remaining two modalities and all other cells use seed 17 only. Cross-seed aggregation is descriptive ( $n=3$ ).

The image-only observation arms, which would test visual state and history channels without the shared textual serialization, are a frozen-protocol proposal with no outcomes and no GPU authorization [TODO: image-only arms, issue #128]. The full-scope v1 generalization and second-backbone branches remain open under their terminal VALID\_STOP receipts [TODO: issues #96/#98/#102/#103]. Release URLs for the published evidence bundles are pending handoff [TODO: release URLs, issue #122].

## 6 DISCUSSION AND LIMITATIONS

The results in Section 5 trace a conditional boundary around learned search execution. Under trusted successors, the trained Search Process Policy produces valid, invariant-compliant traces for certain algorithm families and fails in localizable ways for others. This section interprets those findings, states the runtime boundary that limits every claim, and identifies the threats to validity that bind the current evidence.

**Runtime boundary.** Every positive result depends on the Trusted Search Runtime. The runtime validates emissions, applies transitions, maintains Search Memory, and checks Algorithm Invariants on every operation. It does not choose omitted operands, reorder candidates, or repair invalid emissions. The replay-determinative test attributes each exploration decision to the model. This separation enables clean attribution and equally constrains what the evidence can claim. The contract does not establish that the model conducts unassisted internal search, holds unbounded frontier state, or possesses general planning autonomy. The model issues Typed Search Operations into an external executor. Structural and process competence under this contract means emitting correct operations under external state management. The successor-prediction null confirms the runtime dependence: with model-generated transitions, success drops from 45/45 to 1/45, and structural control collapses.

**What the evidence demonstrates.** The BFWS structural gate provides the single positive learned-control comparison. On its frozen 15-task text-state development panel, process SFT achieved 0.933 success versus 0.467 for random-valid, with a paired bootstrap lower bound of 0.2 (Section 5). This is a development-stage finding from one training seed on one algorithm. It shows that process supervision can train a policy whose full-episode traces exceed the oracle-assisted validity control on the tested panel.

The expanded 24-task baseline marks the boundary. Learned success appears only in the additive best-first cells (125/288 versus 240/288 for random-valid, 0/288 for the base). BFS and width-based best-first score 0/24 in every modality. The pattern is algorithm-specific, not modality-specific. Process SFT succeeds under weighted A\* and greedy best-first regardless of observation modality, and it fails under the queue-discipline algorithms in all three.

Failure-mechanism calibration localizes where learned execution breaks. The base fails on its first decision in every episode (447/447), producing malformed outputs or rejected frontier operations. Learned arms fail later. The dominant learned failure kind is `other_invariant_violation`: parsed, source-correct operations rejected because they violate the declared search invariant (229 of 407 learned failures). Learned first failures cluster in steps 0–2 (137/88/46), while oracle-control failures are all late expansion-budget exhaustion. Process supervision teaches the policy to emit well-formed operations but not to maintain invariant-consistent frontier ordering across a full episode.

The observation-corruption suite identifies which input channel carries the learned control. Masking or shuffling the text channel drops success by up to  $-0.944$ . Blanking or degrading the visual channel costs at most  $-0.111$ , with intervals including zero. The multimodal text-minus-visual isolation contrast is  $-0.694$   $[-0.889, -0.472]$ . Under the frozen observation contract, the learned controller depends on the textual search-memory serialization.

Reduced-scope generalization testing (128/150 under 25 semantics-preserving perturbation variants, versus 0/150 for the base) shows that the learned adapters survive surface-level input variation on the additive best-first family.

**What the evidence does not demonstrate.** Five executed branches returned null or negative results. Model-generated successors collapsed success from 45/45 to 1/45 (Section 5). Two DAGger iterations on BFS added no advantage over exposure-matched continued SFT (1/72 versus 1/72 unseen successes), and DAGger carried the higher invalid-operation rate. Zero-shot transfer to FOLIO, GSM8K, and HumanEval left no comparison surviving Holm correction (0/36, minimum adjusted  $p$  0.8156). The reduced second-backbone evaluation with InternVL3.5-8B reached no goals (0/36 for both base and process SFT). Because InternVL3.5-8B shares the Qwen3-8B LLM family, this contrast isolates the vision tower, connector, image tokenization, and multimodal recipe rather than the language model. Curriculum-by-modality produced no ordering interaction, with all four paired interaction intervals including zero. The matched-modality evidence (process SFT at 5/12 text, 6/12 visual, 6/12 multimodal versus 10/12 random-valid in each modality) does not support an intrinsic modality effect.

**Threats to validity.** *Training-seed coverage.* Most cells rest on a single training seed (17). A prospective replication (issue #127, closed 2026-09-21) measured seed variance for exactly two cells using seeds 17, 29, and 71, with 534 new episodes independently replayed. The headline process-SFT cell (greedy best-first  $\times$  multimodal-state, 24-problem panel) replicated at 21/24, 21/24, 22/24 (mean 0.889, range  $[0.875, 0.917]$ ). The primary positive contrast against the pretrained base was positive at every seed ( $+0.875/+0.875/+0.917$ ) with no sign flip, and the contrast against random-valid stayed negative at every seed ( $-0.125/-0.125/-0.083$ ). The DAGger iteration-1 versus continued-SFT null replicated at every seed (0/72 both arms, paired contrast  $+0.000$   $[0, 0]$ ). The published iteration-2 verdict (1/72 versus 1/72, seed 17 only) is unchanged. Cross-seed aggregation is descriptive ( $n = 3$ ). Cell A tested one modality (multimodal-state), so two of the three headline modalities remain unreplicated. Every other cell remains single-seed. Stability on three seeds in one cell does not guarantee stability elsewhere, and the descriptive aggregation is not a powered significance test.

*Development-stage protocols.* All panels are development panels. The held-out 45-task manifest was frozen before outcomes and has never been accessed [TODO: held-out final primary evaluation]. Preregistration prerequisites for that evaluation remain undecided (Section 4). Until it executes, the BFWs comparison and the algorithm-specific pattern are bounded development-stage findings.

*Saturating controls as bounds.* Random-valid saturates at 100% on BFS and both additive best-first panels. Where it saturates, the learned-to-control gain is unmeasurable rather than zero. The BFWs development panel is the only panel where random-valid falls below 1.0, and it is the only one that reports a positive learned-control comparison. Saturation bounds measurability, not policy quality.

*Text-scaffold confound in the corruption finding.* The corruption result shows that the learned policy depends on the language channel. It does not show that rendered images are inherently uninformative. All three adapters share the same textual search-memory and candidate serialization, so the visual channel is dispensable given that shared scaffold. An image-only observation arm (issue #128) that removes the shared serialization would test whether the visual channel carries indepen-

dent decision-relevant information, but that arm is a frozen-protocol proposal with no outcomes and no GPU authorization [TODO: image-only arms, issue #128].

The full-scope v1 generalization and second-backbone branches remain open under terminal VALID\_STOP receipts [TODO: full-scope v1, issues #96/#98/#102/#103]. Release URLs for evidence bundles are pending handoff [TODO: release URLs, issue #122].

## 7 CONCLUSION

Can vision-language models learn to execute classical search algorithms? The evidence supports a conditional, development-stage answer. On the BFWs structural-gate development panel, process SFT exceeded random-valid (0.933 versus 0.467, bootstrap lower bound 0.2). On the expanded baseline, learned success appeared only in the additive best-first cells, always below saturating controls. BFS and width-based best-first showed zero learned success. Failure-mechanism analysis places learned failures at invariant violations in early decisions, not at malformed outputs. Observation-corruption analysis places the learned control in the textual search scaffold, not in the rendered visual channel. These findings map where execution is learnable under the tested interface. They do not constitute a general efficacy claim.

Several claims remain unsupported. No final comparative efficacy finding exists because the held-out manifest has not been accessed [TODO: held-out final primary evaluation]. Model-generated successors collapse structural control (45/45 to 1/45). DAgger adds no advantage. Transfer is null. A second backbone reaches no goals. The evidence does not establish an intrinsic modality effect or general planning autonomy. Training-seed variance has been measured for two cells (three seeds each, descriptive aggregation). Every other cell is single-seed.

The next gated step is the held-out final primary evaluation on the frozen 45-task manifest, under a preregistered protocol with decided estimands, budget rules, and statistical tests (Section 4). Release URLs for evidence bundles are pending [TODO: release URLs, issue #122].

## REPRODUCIBILITY STATEMENT

Every stage of the study runs under a frozen outcome-blind protocol. Development panels and held-out manifests are frozen before any model outcome. Preregistered gates close each stage with a recorded receipt (PASS, VALID\_STOP, INVALID, or ANCESTOR\_STOP), and no gate was revised after observing results. Training seeds are frozen pre-launch: seed 17 for every cell after the historical pilot, with seeds 29 and 71 added for the two prospective replication cells. No seed was dropped after outcomes. The replay-determinative interface makes every episode independently verifiable from its raw emissions alone. The policy emits Typed Search Operations with explicit operands, and the Trusted Search Runtime validates and applies each operation without filling omitted choices or repairing outputs. Every episode record (model and control) across the seven branches, the observation-corruption suite (360 new + 144 reused bindings), and the #127 amendment (534 episodes) was independently replayed with zero missingness. The 1,152 expanded-baseline bindings include 576 model episodes and 576 CPU control/reference episodes.

Two process corpora supply the training data. The BFS text-state corpus contains 90 replay-verified traces and 25,109 process rows. The BFWs corpus covers 105 tasks and 69,019 replay-verified teacher decisions, projecting to a 47,780-train / 21,239-dev split. Both passed no-overlap, no-conflict, live-parity, and token-budget audits before use. Per-cell training configurations, evaluation manifests, LoRA adapter weights with checkpoint digests, and the synthesis claims inventory are recorded for each executed stage.

The program consumed 68.5244 of a capped 336 GPU-hours on 2×NVIDIA A100 80 GB, with a per-branch ledger that retains failed and cutoff attempts at their recorded cost. Paired contrasts use whole-instance bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals. Transfer comparisons use exact McNemar tests with Holm correction across the 36 adapter-base pairs. Strata with fewer than 8 instances are reported as descriptive only. All seeds, panel memberships, corruption parameters, and frozen protocols are stated in the evaluation manifests. [TODO: release] Public release URLs and artifact DOIs have not yet been assigned and will appear in the final version when the release and handoff are complete.

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## A OPERATION SCHEMAS

Each declared algorithm defines a vocabulary of operation types  $\tau$ . A policy emission  $u_t = (\tau_t, \theta_t)$  pairs a type from that vocabulary with an operand vector  $\theta_t$  carrying every exploration-determining argument. The runtime rejects any emission whose type falls outside the declared vocabulary or whose operand vector omits a required field. Every operand that determines exploration must originate from the policy (Section 3).

**BFS.** The primary operation is node expansion. Its operands name the source state (the current frontier head), the grounded successors selected for insertion, the action producing each successor, and the enqueue position for each. The runtime checks that the named source matches the FIFO frontier head, that each successor results from an applicable grounded action in the source state, that the successor is unvisited, and that each insertion targets the FIFO tail. A goal-test operation names the state to test. The BFS pilot corpus (#111) exposes exact grounded successor candidates and full visited membership at each step.

**BFWS (complete unpruned).** Expansion operands carry the novelty value and goal-count priority of the source state alongside the fields above. Insertion operands carry each candidate’s duplicate membership, novelty, and priority. The runtime checks that the selected source has the declared best priority, that duplicate membership is correctly reported, and that insertions follow the declared ordering. The BFWS corpus (#57–#58) exposes candidate duplicate, novelty, priority, insertion, and enqueue facts at every training observation.

**Additive best-first.** Operands carry the source state’s accumulated path cost  $g$ , heuristic estimate  $h_{\text{add}}$ , and composite value  $f = g + w \cdot h_{\text{add}}$  with  $w = 3$  for the weighted variant, or  $f = h_{\text{add}}$  for the greedy variant. Insertion operands carry per-successor  $g$ ,  $h_{\text{add}}$ , and  $f$ . The runtime checks that the expanded state has the minimum  $f$  in the frontier under the declared tie-breaking rule and that all arithmetic is consistent.

**Validity preconditions.** An emission passes through a fixed sequence of checks. Failure at any step rejects the emission with the corresponding kind and terminates the episode. The checks, in order:

Table 10: Rejection taxonomy from the failure-mechanism calibration (#125). Counts are failures per arm class across 2,910 replay-verified episodes (905 total failures, 0 unclassified). Share is within the arm class.

| Arm class                    | Rejection kind               | Failures | Share  |
|------------------------------|------------------------------|----------|--------|
| Base (447 ep., 447 fail.)    | malformed.output             | 379      | 84.8%  |
|                              | invalid.frontier.operation   | 67       | 15.0%  |
|                              | other.invariant.violation    | 1        | 0.2%   |
| Learned (906 ep., 407 fail.) | other.invariant.violation    | 229      | 56.3%  |
|                              | malformed.output             | 72       | 17.7%  |
|                              | inapplicable.grounded.action | 56       | 13.8%  |
|                              | effect                       | 27       | 6.6%   |
|                              | schema                       | 16       | 3.9%   |
|                              | invalid.frontier.operation   | 5        | 1.2%   |
|                              | source.state.mismatch        | 1        | 0.2%   |
|                              | static.context               | 1        | 0.2%   |
| Oracle control (51 fail.)    | expansion.budget.exhausted   | 51       | 100.0% |

1. **Parse.** The raw output must parse into a well-formed Typed Search Operation with a recognized type and complete operand vector. Failure: `malformed.output`.
2. **Source identity.** The named source state must match the state the algorithm expects. Failure: `source.state.mismatch`.
3. **Action applicability.** Every grounded action in the operands must be applicable in the source state. Failure: `inapplicable.grounded.action`.
4. **Successor schema.** Any named successor state must satisfy the domain schema. Failure: `schema`.
5. **Successor effects.** The named successor must match the declared effects of the applied action. Failure: `effect`.
6. **Static context.** Referenced predicates, objects, and constants must be consistent with the task instance. Failure: `static.context`.
7. **Frontier discipline.** The frontier manipulation must respect the declared ordering (FIFO for BFS, priority for BFWS,  $f$ -order for additive best-first). Failure: `invalid.frontier.operation`.
8. **Algorithm invariant.** Any remaining algorithm-specific constraint not covered above. Failure: `other.invariant.violation`.

The two broad categories are *malformed-output* rejections, where the emission never reaches the operation checker, and *post-parse rejections*, where the emission parses but fails a runtime check. The oracle-assisted random-valid control, which emits only runtime-accepted operations, has a single failure kind: `expansion.budget.exhausted`, recorded when the frozen episode budget runs out before a goal is reached.

**Rejection counts.** Table 10 gives the counts from the failure-mechanism calibration (#125), which classified all 905 failures across 2,910 replay-verified episodes with zero unclassified. The `effect`, `schema`, and `static.context` kinds arise in successor-prediction episodes, where the model generates a dynamic-state successor.

## B ALGORITHM INVARIANT DEFINITIONS

The Trusted Search Runtime checks the following invariants on every operation. Stepwise validity and invariant compliance do not establish termination, completeness, optimality, or episode success (Section 3).

**BFS.** The frontier is a FIFO queue. Every expansion must select the queue head as its source. Every insertion must target the queue tail. A state already in the visited set must not be re-inserted. This invariant is sufficient for completeness in finite state spaces with unit-cost transitions.

**BFS (complete unpruned).** The frontier is a priority queue ordered first by novelty rank (lower is better), then by goal count among states with equal novelty. No generated successor is pruned. A successor whose state already appears in the frontier or visited set is not re-inserted.

**Additive best-first, weighted ( $w = 3$ ).** The frontier is a priority queue ordered by  $f(s) = g(s) + 3 \cdot h_{\text{add}}(s)$ , where  $g(s)$  is the accumulated path cost and  $h_{\text{add}}(s)$  is the additive heuristic. The expanded state must have the minimum  $f$  in the frontier. When a successor is generated with a lower  $g$  than an existing frontier entry for the same state, the entry is reopened with the improved cost.

**Additive best-first, greedy.** The frontier is ordered by  $h_{\text{add}}(s)$  alone, with no path-cost component. The expanded state must have the minimum  $h_{\text{add}}$ . Reopen and duplicate rules follow the same convention as the weighted variant.

## C TRUSTED SEARCH RUNTIME

The runtime provides three guarantees that hold across all declared algorithms.

1. **No-repair property.** The runtime validates, applies, persists to Search Memory, or rejects each emission. It never chooses omitted operands, reorders candidates, silently repairs an invalid emission, or substitutes a default search decision.
2. **Reject-and-terminate.** On the first rejected emission the episode terminates with the rejection kind recorded. There is no retry path.
3. **Replay determinism.** Replaying the raw emissions  $u_{1:T}$  against the same task instance and the same initial Search Memory reproduces the explored trace. Any step whose outcome depends on runtime repair would violate this property.

The procedures below specify the per-step check and the episode loop. No algorithm package is loaded. The listing uses  $\text{\LaTeX}$  tabbing.

```

procedure STEP( $u_t, M_{t-1}, A, x$ )
  // Phase 1: Parse the raw emission
  if  $u_t$  does not parse as  $(\tau, \theta)$  with  $\tau \in \text{vocab}(A)$  then
    return REJECT(malformed_output)
  // Phase 2: Validate source identity
  if  $\theta.\text{source} \neq \text{expected}(A, M_{t-1})$  then
    return REJECT(source_state_mismatch)
  // Phase 3: Validate operands against the task and state
  for each grounded action  $a$  named in  $\theta$  do
    if  $a$  is not applicable in  $\theta.\text{source}$  under  $x.D$  then
      return REJECT(inapplicable_grounded_action)
  for each successor state  $s'$  named in  $\theta$  do
    if  $s'$  violates the domain schema then
      return REJECT(schema)
    if  $s' \neq \text{effects}(a, \theta.\text{source})$  then
      return REJECT(effect)
  if any static-context reference in  $\theta$  is inconsistent with  $x$  then
    return REJECT(static_context)
  // Phase 4: Check Algorithm Invariant
  if frontier discipline of  $A$  is violated by  $(\tau, \theta, M_{t-1})$  then
    return REJECT(invalid_frontier_operation)
  if any other invariant of  $A$  is violated then
    return REJECT(other_invariant_violation)
  // Phase 5: Apply and persist
   $M_t \leftarrow \text{apply}(\tau, \theta, M_{t-1})$ 
  return ACCEPT( $M_t$ )

procedure EPISODE( $x, A, \phi_m, B$ )
   $M_0 \leftarrow \text{InitSearchMemory}(A, x)$ 
  for  $t = 1, 2, \dots, B$  do
```

```

1188    $o_t \leftarrow \phi_m(c_t(x, M_{t-1}))$  // build observation
1189    $u_t \leftarrow \pi(o_t)$  // policy emission
1190    $r \leftarrow \text{STEP}(u_t, M_{t-1}, A, x)$ 
1191   if  $r$  is REJECT( $kind$ ) then
1192     return ( $t$ , FAIL,  $kind$ )
1193    $M_t \leftarrow r.M$ 
1194   if GoalTest( $M_t, x.G$ ) = TRUE then
1195     return ( $t$ , SUCCESS)
1196   return ( $B$ , FAIL, expansion_budget_exhausted)

```

Each call to STEP consumes one unit of the frozen episode budget  $B$ , whether the emission is accepted or rejected. Invalid operations are therefore charged to the budget (Section 4).

## D DATA AND SPLIT PROCEDURES

All training and evaluation splits are by whole task instance. No state, decision, or target from an evaluation task appears in any training row. Every released corpus passed no-overlap, no-conflict, live/training-parity, and token-budget audits before use.

**BFS pilot corpus (#111).** The v6 text-state process corpus holds 25,109 process rows split into 12,994 train and 12,115 dev rows, drawn from 90 replay-verified trusted traces covering 90 distinct tasks. The release assigns splits by semantic task identity, with zero cross-split task overlap. The v8 evaluation panel selects 15 tasks from this corpus’s development set, not its training set. No v8 evaluation-panel task contributes training rows, consistent with the whole-instance rule above. Corpus observations expose the canonical task context, exact grounded successor candidates, and full visited membership. Release audits recorded zero semantic, input, and input-target overlap, zero conflicting identical-input labels, zero teacher-operation rejections, and zero live/corpus mismatches, all within an 8,192-token input cap and a 384-token output allowance.

**BFWS corpus (#57–#58).** The corpus covers the frozen 105-task development panel with 69,019 replay-verified teacher decisions. Its release contains 69,019 process records, 67,215 operational records, and a process projection of 47,780 train and 21,239 dev rows. Each step’s observation exposes candidate duplicate membership, novelty values, goal-count priorities, insertion positions, and enqueue facts. Audit counters recorded zero overlap, mismatch, rejection, or held-out-access violations.

**Evaluation manifests.** The BFS pilot uses a frozen 15-task outcome-blind development panel. The BFWS structural gate froze a 105-task development panel and a separate 45-task held-out manifest. The held-out manifest has never been accessed, and no model prediction, training decision, or analysis references it. The expanded baseline evaluates 24 tasks from 12 domains under 4 algorithms, 3 modalities, and 4 arms (1,152 logical bindings, 576 model episodes, all independently replayed). The curriculum stage evaluates 27 unseen tasks. The generalization-v2 protocol admits 25 derived tasks from 93 eligible semantics-preserving variants (uniform  $k=5$  cheapest-prefix per family, membership frozen before execution).

## E RECEIPT TABLES

Table 11 gives the per-branch gate receipts from the synthesis ledger (#120). Five branches executed to completion at v1 scope. Two stopped at their frozen v1 cost-admission gates with terminal VALID\_STOP receipts and later completed at reduced scope under new v2 protocols. Every branch reconciles with zero missing evidence.

At the #120 synthesis snapshot, the program recorded 135 attempts (96 succeeded, 36 failed, 3 cutoff). Failed and cutoff attempts retain their hours in the ledger. Every attempt ended before the absolute scheduler cutoff of 2026-09-21T11:55:19Z.

The observation-corruption suite (#126) added 2.22 GPU-h to the generalization branch. The training-seed replication addendum (#127, closed 2026-09-21) added 12.79 GPU-h and 534 inde-

Table 11: Per-branch gate receipts for the expanded study program (through #126/#127 addenda). Program spending is 68.52 of the 336 GPU-h cap. Entries and the total are rounded independently from the ledger. One prospective budget transfer (12.11 GPU-h from recovery reserve to second backbone, ratified 2026-09-19) is reflected in the branch caps. The #127 row draws on existing baseline and DAgger caps, not an additional allocation.

| Branch                    | Tickets              | Terminal state | Coverage    | GPU-h / cap        |
|---------------------------|----------------------|----------------|-------------|--------------------|
| Expanded baseline         | #116–#118            | complete       | 1,152/1,152 | 6.53 / 56          |
| DAgger                    | #78–#84              | complete       | 405/405     | 10.91 / 48         |
| Successor prediction      | #85–#89, #99         | complete       | 90/90       | 10.85 / 64         |
| Curriculum by modality    | #119                 | complete       | 567/567     | 11.78 / 48         |
| Transfer                  | #104–#107            | complete       | 39/39       | 3.60 / 24          |
| Second backbone           | #101–#103, #123      | v2 complete    | 144/144     | 6.03 / 52.11       |
| Generalization/robustness | #96, #98, #124, #126 | v2 complete    | 1,200 + 360 | 6.04 / 32          |
| Recovery reserve          | –                    | reduced        | –           | 0.00 / 11.89       |
| Training-seed replication | #127                 | complete       | 534/534     | 12.79 / –          |
| <b>Total</b>              |                      |                |             | <b>68.52 / 336</b> |

Table 12: Seed-variance replication for Cell A (headline process-SFT cell). Contrasts are paired whole-instance bootstrap (24 tasks, 10,000 resamples, seed 1729). The primary contrast (SFT – base) is positive at every seed with no sign flip.

| Seed | Successes / 24 | SFT – base | SFT – random-valid |
|------|----------------|------------|--------------------|
| 17   | 21/24          | +0.875     | –0.125             |
| 29   | 21/24          | +0.875     | –0.125             |
| 71   | 22/24          | +0.917     | –0.083             |

pendently replayed episodes covering two cells. Together these addenda brought the program total to 68.52/336 GPU-h. See Section F for the seed-variance results.

## F STATISTICS

**Default contrast procedure.** Contrasts are paired by whole task instance. Paired differences use a bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals. Under one training seed and panels of 24 or fewer instances, these intervals are descriptive bounds. They do not support broad superiority claims.

The BFWS gate (#59) used confidence level 0.95 with 10,000 paired whole-instance resamples (bootstrap seed 1729). Its reported lower bound is the 0.025 quantile, i.e., the lower endpoint of a two-sided 95% percentile interval.

**Tiny-strata rule.** Strata with fewer than 8 episodes are reported as descriptive only. No intervals are computed over such strata. They are listed explicitly and never silently pooled or dropped.

**Transfer tests.** Transfer comparisons use McNemar exact tests against the per-example base predictions, with Holm correction across all 36 adapter-by-benchmark comparisons. The minimum adjusted  $p$  in the executed transfer branch is 0.8156. None survives correction.

**Seed-variance addendum (#127).** A prospective training-seed replication measured variance for exactly two cells. Seeds 17, 29, and 71 were frozen before any seed-29 or seed-71 artifact existed. All 534 new episodes were independently replayed. Cross-seed aggregation is descriptive ( $n = 3$  seeds).

Cell A is the headline process-SFT cell (*best\_first\_add\_greedy*  $\times$  *multimodal-state*) evaluated on the 24-problem panel. Each seed trained a fresh LoRA adapter with new initialization and dropout randomness on the identical frozen 512-record membership. Table 12 gives the per-seed results.

Cell set B covers DAgger iteration-1 versus the exposure-matched continued-SFT control under BFS across all three modalities. The null replicates: 0/72 unseen successes for every arm and every seed. The paired contrast (DAgger minus continued SFT) is +0.000 [0.000, 0.000] at each seed.

This addendum supersedes the blanket single-training-seed boundary for these two cells only. Every other cell in the program remains single-seed (seed 17) evidence.

## G MODALITY ADAPTERS AND OBSERVATION CONTRACT

Three fixed Modality Observation adapters are declared. Each maps a decision context  $c_t$  (the canonical task context plus the decision-relevant view of Search Memory  $M_t$ ) to an observation  $o_t = \phi_m(c_t)$  presented to the policy. The adapter is fixed for an entire training or evaluation cell. The adapter contract requires decision sufficiency:  $o_t$  must expose every fact the exact teacher uses to resolve step  $t$  under the declared algorithm.

**text-state.** The state, goal, and search context are serialized as a compact relational representation. Object identities, predicate names, and all Search Memory contents (frontier, visited set, algorithm-specific state) appear as structured text.

**visual-state.** The state and goal are rendered as a pair of  $128 \times 128$  pixel images: a rendered state image and a partial-goal constraint image. Object labels are not embedded in the rendered image. Search Memory contents and the grounded candidate list remain in the textual serialization, even in this adapter.

**multimodal-state.** The observation combines the text-state serialization and the visual-state rendered images. The policy receives the relational text and the image pair jointly.

**Text-scaffold caveat.** Search Memory contents and the candidate list are serialized as structured text in all three adapters, including visual-state. Rendered images carry the planning state and goal visuals, but search bookkeeping (frontier, visited set, novelty counters, candidate list) always travels through the text channel. The observation-corruption suite (#126) confirms this empirically: corrupting the text channel drops success by up to  $-0.944$ , while corrupting the visual channel moves success by at most  $-0.111$  with intervals that include zero.

**Information-loss boundaries.** The 128-pixel unlabelled rendered views can lose symbolic identities. The P3 name-compression family in the generalization-v2 suite is information-lossy: 24/24 variants are text-lossy, 18/24 are multimodal-lossy, and 0/24 are visual-lossy. Modality matching between training and evaluation ensures identical observation format, not information parity. Static-context and goal fields of the observation contain text in every adapter. The available evidence does not establish semantic parity across adapters or an intrinsic modality effect.

## H RETAINED-INFRASTRUCTURE FEASIBILITY NOTES

The terms below describe verified infrastructure components that support corpus generation, evaluation, and evidence production. They constitute feasibility evidence for the runtime and rendering pipeline. They cannot support a Search Process Policy efficacy claim.

**Plan pipeline.** **Plan Submission** transmits a supplied plan from the project adapter to a planning or rendering backend. **Plan Interpretation** converts the submitted plan into the backend’s ordered action representation. An **Action Sequence** is the canonical identity of a supplied plan as a normalized, ordered sequence of grounded action predicates. Casing and whitespace are not part of its identity.

**Rendering pipeline.** **Render Production** creates a visualisation output and its rendered image for a planning state. **Render Validation** confirms that the produced visualisation and image satisfy declared structural and semantic checks. **Plan Provenance** links a produced visualisation to the exact supplied plan that the backend interpreted.

**Certification and evidence.** **Integration Certification** is the finding that Plan Submission, Plan Interpretation, Plan Provenance, Render Production, Render Validation, and required execution-isolation checks all passed. An **Attempt** is one authorized, immutable execution with its own identity and retained outcome, whether successful or not. An **Evidence Bundle** is the self-contained records from an Attempt that allow an independent reproducer to verify its claims offline. Issue #110 supplies deterministic compact delta/event evidence with semantic replay while preserving search semantics, serving the replay-determinism requirement of the Trusted Search Runtime.

**Historical terms.** Planning Certificate, Joint Action-and-Certificate SFT, Adaptive Scaffolding, Support Route, Live Memory, Verified Joint Step, and Route Label are historical CGAS terms. They describe retained infrastructure internals. They cannot support any current Search Process Policy efficacy claim and are not referenced elsewhere in this manuscript.